

Practical - 3

Difference between Heliophytes and Sciophytes

Heliophytes	Sciophytes
<u>Morphological features</u>	
1. Growth of the shoot reduced and internodes remain short.	1. Growth of the shoot is normal and internodes are elongated.
2. Thick stems with well developed mechanical and conducting tissues.	2. Thinner stems with weakly developed mechanical and conducting tissue.
3. Very well branched system	3. Branching less
4. Roots numerous, longer and branched	4. Roots much less in number.
5. Leaves narrow, linear thicker and of hard texture	5. Leaves large broad and long thin.
6. Leaves, often directed sharply or vertically upwards.	6. Leaves are horizontal.
7. Leaves may be folded	7. Leaves are usually flat and smooth.
8. Leaves are usually isobilateral	8. Leaves are always dorsiventral
9. Leaves glossy and good reflector of light.	9. Leaves often velvety beset with refractive papillae which serve to collect the obliquely ray of light.

10. Leaves covered with hairs

10. Leaves less hairy and glabrous

Anatomical

Features

1. Cuticle as well as epidermis is thick.

1. Cuticle as well as epidermis is thin.

2. Upper epidermis lacks chlorophyll

2. Upper epidermis may contain chlorophyll

3. Lateral walls of epidermal cells less wavy.

3. Lateral walls of epidermal cells more sinuous.

4. Stomata smaller, closer and usually on the lower surface

4. Stomata larger and present on both the surfaces.

5. Mesophyll cells smaller.

5. Mesophyll cells larger

6. Well developed palisade parenchyma may be on both sides.

6. Palisade parenchyma poorly developed.

7. Spongy tissue weakly developed with small intercellular spaces

7. ~~Spongy tissue well developed with large intercellular spaces~~

8. Chloroplasts fewer and smaller

8. Chloroplasts numerous and large

Physiological

Features

1. Photosynthetic rate lower per unit area

1. Photosynthetic rate higher per unit area.

2. Respiration rate as well as transpiration rate are higher

2. Respiration rate as well as transpiration rate is lower

3. Flowering earlier

4. Osmotic pressure higher due to higher salt and sugar content

3. Flowering delayed.

4. Osmotic pressure lower

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